AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2019

Chemistry Paper II

Time: 2 hours 10 minutes Marks: 50

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are ELEVEN questions. Answer ALL questions. Questions 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 4 Marks)

a. Describe Heisenberg's uncertainty principle. (1 Mark)

_____

_____

b. Identify each principle/ rule of electronic configuration described in the given table.
(3 Marks)

S. No.	Description	Name of Principle/ Rule
1	Electron should be filled in energy subshells in order of increasing energy values.	
2	Two electrons in the same orbital should have opposite spin.	
3	Each orbital in a sublevel is separately occupied before any orbital is doubly occupied.	

Q.2. (Total 4 Marks)

Describe the concept of conjugate acids and bases using the following reactions as examples.

a. Reaction between hydrogen chloride and water (2 Marks)

_____

_____

_____

_____

b. Reaction between ammonia and water (2 Marks)

_____

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Q.3. (Total 3 Marks)

a. Arrange the following bonded atoms in the correct order of decreasing bond energy.

C≡C, C≡O, C≡N (1 Mark)

b. Give a justification for your arrangement in part a. (2 Marks)

Q.4. (Total 3 Marks)

Give a reason why

a. ionic compounds are soluble in water but insoluble in non-aqueous solvents. (2 Marks)

b. some covalent compounds dissolve in water. (1 Mark)

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Q.5. (Total 4 Marks)

a. State Dalton's law of partial pressure. (1 Mark)

b. Explain why it is dangerous for divers to hold their breath during scuba diving. (3 Marks)

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Q.6. (Total 4 Marks)

a. Describe the role of a catalyst in a reversible reaction. (2 Marks)

b. Identify the salt from the given table which has the

S. No.	Salts	Ionic Product	K_{sp}
1	Ag_2CO_3	$[Ag^+]^2\ [CO_3^{2-}]$	8.1×10^{-12}
2	AgCl	$[Ag^+]\ [Cl^-]$	1.8×10^{-10}
3	$Fe(OH)_3$	$[Fe^{3+}]\ [OH^-]^3$	1.6×10^{-39}

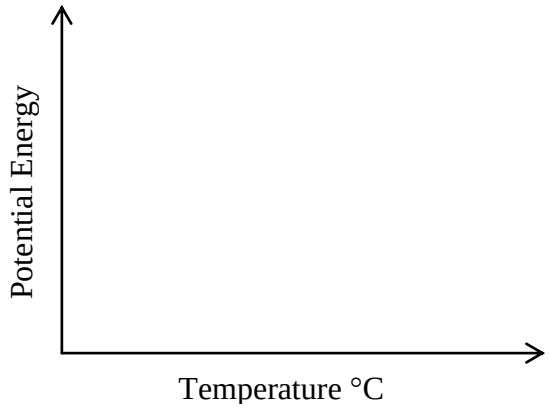
i. highest tendency to dissociate. (1 Mark)

ii. least tendency to dissociate. (1 Mark)

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Q.7. (Total 4 Marks)

a. Draw a labelled energy diagram for an exothermic reaction. (2 Marks)

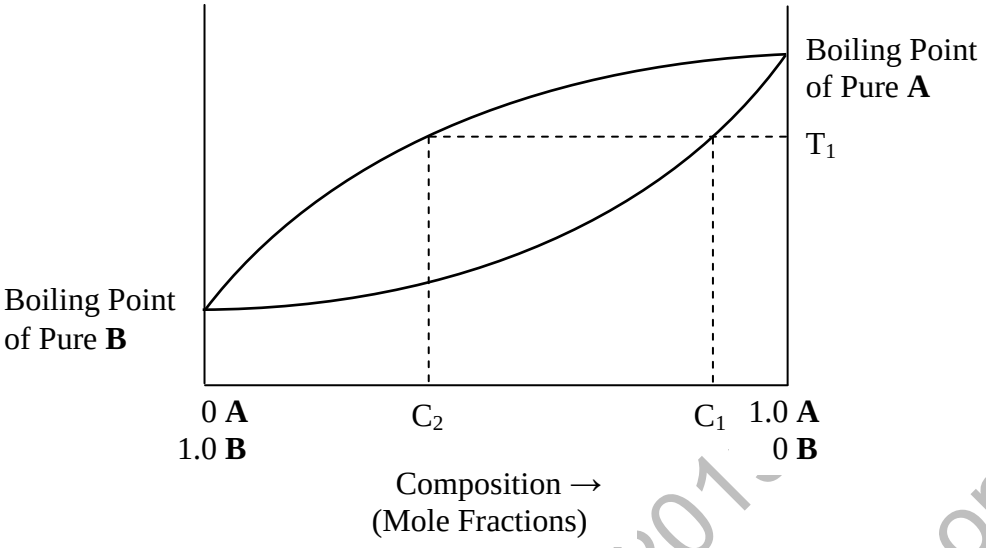


b. Describe the effective collision among molecules with reference to collision theory. (2 Marks)

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Q.8. (Total 4 Marks)

Consider the composition-temperature graph of an ideal solution of two miscible liquids **A** and **B**.



a. Which liquid is more volatile, **A** or **B**? Give a reason for your identification. (2 Marks)

b. Interpret the following parts in the given graph.

i. Point C_1 (1 Mark)

ii. Point C_2 (1 Mark)

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Q.9. (Total 5 Marks)

Use the given information to answer the following questions.

$$Pt_{(s)} \left| MnO^{-}_{4(aq)}, H^{+}_{(aq)}, Mn^{2+}_{(aq)} \right\| Sn^{4+}_{(aq)}, Sn^{2+}_{(aq)} \left| Pt_{(s)} \right.$$

The half-cell reactions and potentials are:

Equation 1 $MnO^{-}_{4(aq)} + 8H^{+}_{(aq)} + 5e^{-} \rightarrow Mn^{2+}_{(aq)} + 4H_2O_{(l)}$ $E° = +1.51V$

Equation 2 $Sn^{4+}_{(aq)} + 2e^{-} \rightarrow Sn^{2+}_{(aq)}$ $E° = +0.15V$

a. Which equation, 1 OR 2, is the oxidation half-cell reaction? Show the correct representation of the identified equation with its cell potential. (2 Marks)

b. Write a balanced chemical equation for the overall cell reaction that occurs on the conduction of current. (2 Marks)

c. Calculate the standard potential (E°) of the given electrochemical cell. (1 Mark)

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Q.10. (Total 7 Marks)

EITHER

a. The combustion analysis of an organic compound with relative molar mass 99 g shows that it contains 4.04% hydrogen, 24.24% carbon and 71.72% chlorine.

(**Note**: Atomic mass of H = 1 amu, C = 12 amu and Cl = 35.5 amu)

What are the (i) empirical formula and (ii) molecular formula of the given compound? Show the steps of working clearly.

OR

b. Bohr's equation for the calculation of energy associated with electron in the nth orbit of hydrogen atom is $E_n = \frac{-me^4}{8\varepsilon_\circ^2 n^2 h^2}$ where $\frac{-me^4}{8\varepsilon_\circ^2 h^2} = -2.18 \times 10^{-18}$ J.

Justify that the energy difference $E_3 - E_2 > E_4 - E_3$. Show the steps of working and determine the number of times $E_3 - E_2$ is greater than $E_4 - E_3$.

(**Note**: Write answers up to two decimal places.)

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Q.11. (Total 8 Marks)

EITHER

a. The given reaction takes place in an acidic medium.

$$PbO_{2(s)} + I^-_{(aq)} \rightarrow Pb^{2+}_{(aq)} + I_{2(g)}$$

Balance the given equation using the half-reaction method.

(**Note**: Show all the steps of balancing in sequence.) (8 Marks)

OR

b.

Change in Standard Enthalpy	Value/ kJ mol^{-1}
ΔH°_{at} Heat of atomisation of $Ca_{(s)}$	+ 179
ΔH°_{BDE} Bond dissociation enthalpy of $F_{2(g)}$	+ 158
ΔH°_{i} First ionisation energy of $Ca_{(g)}$	+ 590
ΔH°_{i} Second ionisation energy of $Ca^+_{(g)}$	+ 1,150
ΔH°_{ea} First electron affinity of $F_{(g)}$	− 328
ΔH°_{f} Heat of formation of $CaF_{2(s)}$	− 1,220

Using the given data,

i. construct a labelled Born-Haber cycle for calcium fluoride (CaF_2). (6 Marks)

ii. calculate the lattice enthalpy of CaF_2. (2 Marks)

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END OF PAPER

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